12.3 MACHINING CHARGES

A part or component obtained only after it is subjected to several operations performed on different machine tools. The machining charges are calculated by the duration of machining on each of these machine tools involved in manufacturing the particular component. The machining charges differ for different type of machine tools.

The table prepared here is approximately for calculating the machinery charges. The charges may vary according to the cost of the machine, capacity and accuracy of the machine tool.

MACHINE	DURATION	RATE PER HOUR
Drilling machine	1 hour	₹100.00
Lathe	1 hour	₹150.00
Shaping machine	1 hour	₹200.00
Milling machine	1 hour	₹250.00
Grinding machine	1 hour	₹300.00

12.4 WAGES FOR THE WORKER

This includes the amount of wages paid to the workers employed for operating the machine tools and the expenditure for miscellaneous operations performed on the workpieces. Charges for hacksaw cutting, filing, fitting, and assembling are counted under this head.

12.5 COST FOR MAKING ACCESSORIES LIKE JIGS AND FIXTURES

If the production process involves the usage of jigs and fixtures, (used to hold and guide the job) the cost paid for manufacturing or purchasing the same should be taken into account.

12.6 ADMINISTRATIVE EXPENDITURE

The administrative expenditure can also be called as 'overhead charges'. It includes:

1. Depreciation (wear and tear on machine tools and other tools)
2. Repairs and maintenance of the machine tools.
3. Expenditure on fuels and electricity.
4. Rent and maintenance paid for the building of the manufacturing unit.
5. Salary paid to the administrative staff and expenses for purchasing office records.
6. Salary paid to the supervisors, storekeepers and watchman.
7. Expenditure for advertising, travel and postage.
8. Amount spend for the Welfare of the workers, and contribution to the employee's fund (it includes medical insurance and their child education fund)
9. Interest on capital.

12.7 PROFIT

Profit is calculated in the range of 10% to 30% of the total cost of estimation.

The profit percentage is varies for each factory according to their capital invested.

Example Problems

100 Nos. of spur gears are required with the following specifications.

No. of teeth	=	60
Module	=	3 mm
Width of the gear	=	30 mm
Diameter of the stock	=	40 mm

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